PROBABILITY: MORE SAMPLE SPACE DIAGRAMS **14/10/25**

1. A maths teacher uses the following model: When a Fifth former arrives at a Maths lesson the probability that he will be late is $\frac{2}{9}$ and the probability that he will have forgotten his prep is $\frac{1}{4}$. The probability that he will be on time and have remembered his prep is $\frac{7}{12}$.

   According to this model: what is the probability that a Fifth former:
   a. will be late or have forgotten his prep (or both);
   b. will be on time;
   c. will have remembered his prep;
   d. remembers his prep given that he is on time;
   e. is on time given that he remembers his prep?

2. In a video game, players who rank highly in an event have the chance to win an exclusive 'skin' for their avatar. There are three different designs: the Lance (attack boost), Shield (defence boost), and Tractor (production boost), which come in three different colours: Gold (2.5% boost), Red (2% boost) and Blue (1.5% boost). Let the events L, S, T correspond to getting the design Lance, Shield, or Tractor, respectively, and let the events G, R, B correspond to obtaining the colour Gold, Red, or Blue respectively.

   You are given the following information:

   $P(L) = 0.33$, $P(G) = 0.1$, $P(B) = P(G \cup R)$, $P(L \cap R) = 5P(S \cap G)$, and
   $P(S \cap B) = P(T \cap B) = P(T \cap R) = P(S \cap R) = 3P(T \cap G)$.

   Find $P(L \cap G)$.

3. A simplified model of the human blood-type system has four blood types: A, B, AB, and O. There are two antigens: anti-A and anti-B that react with a person's blood in different ways depending on the blood type. Anti-A reacts with blood types A and AB, but not with B or O. Anti-B reacts with blood types B and AB but not with A or O. Suppose that a person's blood is sampled and tested with the two antigens. Let A be the event that the blood reacts with anti-A, and let B be the event that the blood reacts with anti-B.

   Classify the person's blood type using the events A, B, and their complements.

4. In the situation of Question 6, suppose that for a randomly chosen person the probability that they have type O blood is 0.5, the probability of type A blood is 0.34, the probability of type B blood is 0.12.
   a. Find the probability that each of the antigens will react with the person's blood.
   b. Find the probability that both of the antigens will react with the person's blood.

Bonus Questions

5. For events A and B it is known that $P(A) = 2/3$, $P(A \cup B) = 3/4$ and $P(A \cap B) = 5/12$. Find $P(B)$ and illustrate this information on a Venn diagram.

6. For events C and D, $P(C) = 0.7, P(D \cup C) = 0.9, P(C \cap D) = 0.3$. Find
   a. $P(D)$ b. $P(D' \cap C)$ c. $P(D \cap C')$ d. $P(D' \cap C')$

7. Using the rule that $P(A \cup B) = P(A) + P(B) - P(A \cap B)$, show that for any three events A, B and C:

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(C \cap A) + P(A \cap B \cap C)$$